ChatGPT API Product Listing Generator

Final Lab Report

1. How the API Integration Works

The goal of this project was to automate the creation of e-commerce product listings using the OpenAI API and product images. The application loads product metadata from the dataset, including the product ID, product name, price and category. Each corresponding product image is converted into a Base64-encoded string so that it can be included in the API request.

For each product, the program creates a structured prompt instructing the AI to analyse the image and generate a product title, detailed description, key features and SEO keywords. The text prompt and encoded image are sent together to the OpenAI API. The returned response is cleaned and converted from JSON text into a Python dictionary using json.loads(). Each successful result is added to a list, and all generated listings are finally saved in product_listings.json.

2. Challenges Faced

One challenge was processing the API response as JSON. Although the prompt requested JSON, the model initially returned the response surrounded by Markdown formatting such as ```json. This caused json.loads() to return the error: "Expecting value: line 1 column 1 (char 0)". I solved this by cleaning the response before parsing it and removing Markdown code fences when present.

Another challenge was preparing the product data correctly. The product information was loaded from styles.csv, while the images had to be matched to the correct product IDs. I therefore constructed each image path using the product ID. I also corrected the product-data processing so that the actual price values from the dataset were included in the generated JSON output.

I also implemented try/except error handling. When the program was tested with an invalid image path, the error was caught and displayed while the program continued processing the remaining products instead of crashing.

3. Quality of the Generated Listings

Overall, the generated listings were well structured and suitable as a starting point for e-commerce content. The model consistently generated titles, descriptions, feature lists and SEO keywords. The regenerated JSON also contains the correct product prices alongside the product ID, original name and category.

The model made good use of visually apparent characteristics such as colour, garment type and visible design elements. For example, the generated listings distinguish between different colours, graphics, tops, skirts, leggings, sweatshirts and jackets.

However, some generated statements were more specific than the supplied metadata or an image alone could reliably establish. Examples included claims about fabric composition, breathability, durability, moisture-wicking technology and machine washability. These details should be verified against trusted product specifications before publication.

4. Potential Improvements

The application could be improved by providing the model with verified product specifications in addition to the image. The prompt could also explicitly instruct the model not to infer materials, technical features or care instructions unless they are supplied in the metadata or clearly visible.

The output could be made more reliable by using stricter structured-output validation instead of relying only on cleaning model-generated JSON. Additional error handling could include retry logic for temporary API failures and validation for missing or unsupported images.

The application could also validate product metadata before sending it to the API. For a production system, additional verified information such as material composition, available sizes, care instructions and inventory data could be supplied to improve the accuracy and usefulness of the generated listings.

Conclusion

The project successfully demonstrates an end-to-end AI API workflow: loading product data, processing images, constructing prompts, calling a multimodal AI model, parsing responses, handling errors and saving structured results. The generated listings show that AI can substantially reduce the manual work involved in creating e-commerce content. The corrected output also demonstrates successful integration of product prices from the dataset. However, human review and verified product metadata remain important because image-based AI can generate plausible product details that are not necessarily confirmed by the source data.